

Q1

k	0	1	2	3	4	5
$P(k)$	$\frac{1}{32}$	$\frac{12}{32}$	$\frac{11}{32}$	$\frac{5}{32}$	$\frac{2}{32}$	$\frac{1}{32}$

 k = no. of times head occur consecutively

Now expectation

$$= \sum xP(k) = (-1) \times \frac{1}{32} + (-1) \times \frac{12}{32} + (-1) \times \frac{11}{32} + 3 \\ \times \frac{5}{32} + 4 \times \frac{2}{32} + 5 \times \frac{1}{32} = \frac{1}{8}$$

Q2

$$\text{Var}(1, 2, \dots, n) = 10$$

$$\Rightarrow \frac{1^2 + 2^2 + \dots + n^2}{n} - \left(\frac{1 + 2 + \dots + n}{n} \right)^2 = 10$$

$$\Rightarrow \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2} \right)^2 = 10$$

$$\Rightarrow n^2 - 1 = 120 \Rightarrow n = 11$$

$$\text{Var}(2, 4, 6, \dots, 2m) = 16 \Rightarrow \text{var}(1, 2, \dots, m) = 4$$

$$\Rightarrow m^2 - 1 = 48 \Rightarrow m = 7 \Rightarrow m + n = 18$$

Q3

$$\text{mean} = \bar{x} = \frac{3 + 7 + 9 + 12 + 13 + 20 + x + y}{8} = 10$$

$$\Rightarrow x + y = 16 \dots (i)$$

$$\text{variance } \sigma^2 = \frac{\sum (x_i)^2}{8} - (\bar{x})^2 = 25$$

$$\frac{9 + 49 + 81 + 144 + 169 + 400 + x^2 + y^2}{8} - 100 = 25$$

$$\Rightarrow x^2 + y^2 = 148 \dots (ii)$$

$$(x+y)^2 = (16)^2 \Rightarrow x^2 + y^2 + 2xy = 256 \Rightarrow xy = 54$$

Q4

If each observation is multiplied with p
and then q is subtracted

$$\text{New mean } \bar{x}_1 = p\bar{x} - q$$

$$\Rightarrow 10 = p(20 - q) \dots (1)$$

and new standard deviations

$$\sigma_2 = |p| \sigma_1 \Rightarrow 1 = |p| (2) \Rightarrow |p| = \frac{1}{2} \Rightarrow p = \pm \frac{1}{2}$$

$$\text{If } p = \frac{1}{2}$$

then $q = 0$ (from equation (1))

$$\text{If } p = -\frac{1}{2}$$

$$q = -20$$

Q5

$$\frac{\sum x_i}{20} = 10 \quad \dots (i)$$

$$\frac{\sum x_i^2}{20} - 100 = 4 \quad \dots (ii)$$

$$\sum x_i^2 = 104 \times 20 = 2080$$

$$\text{Actual mean} = \frac{200 - 9 + 11}{20} = \frac{202}{20}$$

$$\begin{aligned} \text{Variance} &= \frac{2080 - 81 + 121}{20} - \left(\frac{202}{20} \right)^2 \\ &= \frac{2120}{20} - (10.1)^2 = 106 - 102.01 = 3.99 \end{aligned}$$

Q6

$$\text{Mean } (x_i - 5) = \frac{\sum (x_i - 5)}{10} = 1$$

$$\therefore \lambda = \{\text{Mean } (x_i - 5)\} + 2 = 3$$

$$\mu = \text{var } (x_i - 5) = \frac{\sum (x_i - 5)^2}{10} - \frac{\sum (x_i - 5)}{10} = 3$$

Q7

$$\therefore \bar{x} = \frac{\sum_{r=1}^{17} r}{17} = \frac{17 \times 18}{17 \times 2} = 9$$

$$\therefore \bar{y} = ax + b = 17 \Rightarrow 9a + b = 17 \quad \dots (1)$$

$$\therefore \text{Var}(x) = \frac{\sum_{r=1}^{17} r^2}{17} - (\bar{x})^2 = \frac{17 \times 18 \times 35}{17 \times 6} - 9^2 = 24$$

$$\text{and } \text{Var}(y) = a^2 \cdot \text{Var}(x) \Rightarrow 24a^2 = 216 \Rightarrow a = 3 \quad \dots (2)$$

$$\text{Clearly } b = -10$$

$$\text{hence } a + b = -7$$

Q8

$$\text{Variance} = \frac{\sum_{i=1}^{11} b_i^2}{11} - \left(\frac{\sum_{i=1}^{11} b_i}{11} \right)^2$$

$$= \frac{\sum_{r=0}^{10} (b_1 + rd)^2}{11} - \left(\frac{\sum_{r=0}^{10} (b_1 + rd)}{11} \right)^2$$

$$= \frac{11b_1^2 + 2b_1d\left(\frac{10 \times 11}{2}\right) + d^2\left(\frac{10 \times 11 \times 21}{6}\right)}{11} -$$

$$\Rightarrow \text{variance} = 90$$

$$\Rightarrow 10d^2 = 90$$

$$\Rightarrow d = 3$$

Q9

If variate varies from a to b then variance

$$\text{var}(x) \leq \left(\frac{b-a}{2} \right)^2$$

$$\Rightarrow \text{var}(x) < \left(\frac{10-0}{2} \right)^2$$

$$\Rightarrow \text{var}(x) < 25$$

$$\Rightarrow \text{standard deviation} < 5$$

Clearly standard deviation can't be 6.

Q10

Here $n = 10$

$$\begin{aligned}\text{So } S.D. &= \sqrt{\frac{\sum (x_i - p)^2}{n} - \left(\frac{\sum x_i - p}{n}\right)^2} \\ &= \sqrt{\frac{9}{10} - \frac{9}{100}} = \sqrt{\frac{90-9}{100}} = \frac{9}{10}\end{aligned}$$

Q11

Let the two remaining observations be x and y .

$$\therefore \bar{x} = 10 = \frac{5+7+10+12+14+15+x+y}{8}$$

$$\Rightarrow x + y = 17 \dots (1)$$

$$\begin{aligned}\text{var}(x) &= 13.5 = \frac{25+49+100+144+196+225+x^2+y^2}{8} - (10)^2 \\ x^2 + y^2 &= 169 \dots (2)\end{aligned}$$

From (1) and (2)

$$(x, y) = (12, 5) \text{ or } (5, 12)$$

$$\text{So } |x - y| = 7$$

Q12

$$\begin{array}{|c|c|c|c|} \hline \mathrm{x}_i & \mathrm{f}_i & \mathrm{x}_i \mathrm{f}_i & \mathrm{x}_i^2 \mathrm{f}_i \\ \hline 15 & 25 & 375 & 5625 \\ 25 & 35 & 875 & 21875 \\ \hline \end{array}$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{30+70+25x}{4+x} = 25$$

$$\text{Given } \sigma^2 = 50 = \frac{\sum f_i x_i^2}{\sum f_i} - (\bar{x})^2$$

$$\Rightarrow 50 = \frac{450+625x+2450}{4+x} - 625$$

$$\Rightarrow 675 = \frac{2900+625x}{4+x}$$

$$\Rightarrow 50x = 200$$

$$\therefore x = 4$$

Q13

Let two remaining observations are x, y

$$\text{So } \bar{x} = \frac{2+4+10+12+14+x+y}{7} = 8 \text{ (given)}$$

$$\Rightarrow x + y = 14$$

$$\text{Now also } \sigma^2 = \frac{\sum x_i^2}{N} - \left(\frac{\sum x_i}{N}\right)^2 = 16 \text{ (given)}$$

$$= \frac{4+16+100+144+196+x^2+y^2}{7} - 64 = 16$$

$$\Rightarrow 460 + x^2 + y^2 = (16 + 64) \times 7$$

$$\Rightarrow x^2 + y^2 = 100$$

$$\text{Now } (x+y)^2 = x^2 + y^2 + 2xy \Rightarrow xy = 48 \dots$$

$$\text{Now } (x-y)^2 = (x+y)^2 - 4xy = 196 - 192 = 4$$

$$\Rightarrow x - y = 2 \Rightarrow |x - y| = 2$$

Q14

$$\text{Mean} = \frac{3+5+7+a+b}{5} = 5 \Rightarrow a + b = 10$$

$$\text{Variance} = \frac{3^2+5^2+7^2+a^2+b^2}{5} - (5)^2 = 4$$

$$\Rightarrow a^2 + b^2 = 62$$

$$\Rightarrow (a+b)^2 - 2ab = 62$$

$$\Rightarrow ab = 19$$

So a and b are the roots of the equation

$$x^2 - 10x + 19 = 0$$

Q15

$$\begin{aligned}\text{Standard deviation} &= \sqrt{\frac{\sum x_i^2}{N} - \left(\frac{\sum x_i}{N}\right)^2} \\ &= \sqrt{\frac{\sum (x_i - a)^2}{N} - \left(\frac{\sum (x_i - a)}{N}\right)^2} \\ &= \sqrt{\frac{na}{n} - \left(\frac{n}{n}\right)^2} \\ &= \sqrt{a - 1}\end{aligned}$$

Q16

$$\begin{aligned}\text{Mean} &= \frac{\sum x_i f_i}{\sum f_i} \\ &= \frac{0 \cdot {}^nC_0 + 2 \cdot {}^nC_1 + 2^2 \cdot {}^nC_2 + \dots + 2^n \cdot {}^nC_n}{{}^nC_0 + {}^nC_1 + \dots + {}^nC_n}\end{aligned}$$

For finding sum of numerator consider

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n \dots (1)$$

Put $x = 2 \Rightarrow 3^n - 1 = 2 \cdot {}^nC_1 + 2^2 \cdot {}^nC_2 + \dots + 2^n \cdot {}^nC_n$

For sum of denominator

Put $x = 1$ in (1)

$$2^n = {}^nC_0 + {}^nC_1 + \dots + {}^nC_n$$

$$\frac{3^n - 1}{2^n} = \frac{728}{2^n} \Rightarrow 3^n = 729 \Rightarrow n = 6$$

Q1 JEE Main 2020 - 7 Jan (Morning)

An unbiased coin is tossed 5 times. Suppose that a variable X is assigned the value k when k consecutive heads are obtained for $k = 3, 4, 5$ otherwise X takes the value -1 . Then the expected value of X , is :

- (A) $\frac{1}{8}$
- (B) $-\frac{1}{8}$
- (C) $\frac{3}{8}$
- (D) $-\frac{3}{8}$

Q2 JEE Main 2020 - 7 Jan (Morning)

If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then $m + n$ is equal to.....

Q3 JEE Main 2020 - 7 Jan (Evening)

If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then $x \cdot y$ is equal to

Q4 JEE Main 2020 - 8 Jan (Morning)

The mean and the standard deviation (s, d) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q , where $p \neq 0$ and $q \neq 0$. If the new mean and new s, d . become half of their original values, then q is equal to :

- (A) -10
- (B) -20
- (C) -5
- (D) 10

Q5 JEE Main 2020 - 8 Jan (Evening)

The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is :

- (A) 3.99
- (B) 3.98

(C) 4.01

(D) 4.02

Q6 JEE Main 2020 - 9 Jan (Morning)

For observations x_i given $\sum_{i=1}^{10} (x_i - 5) = 10$ and $\sum_{i=1}^{10} (x_i - 5)^2 = 40$. If mean and variance of observations $(x_1 - 3), (x_2 - 3), \dots, (x_{10} - 3)$ is λ & μ respectively then ordered pair (λ, μ) is

(A) (3, 3)

(B) (1, 3)

(C) (3, 1)

(D) (1, 1)

Q7 JEE Main 2020 - 2 Sep (Morning)

Let $X = \{x \in N : 1 \leq x \leq 17\}$ and $Y = \{ax + b : x \in X \text{ and } a, b \in R, a > 0\}$. If mean and variance of elements of Y are 17 and 216 respectively then $a + b$ is equal to :

(A) 7

(B) -27

(C) 9

(D) -7

Q8 JEE Main 2020 - 2 Sep (Evening)

If the variance of the terms in an increasing A.P. $b_1, b_2, b_3, \dots, b_{11}$ is 90, then the common difference of this A.P. is

Q9 JEE Main 2020 - 3 Sep (Morning)

For the frequency distribution:

Variate (x) : $x_1 \quad x_2 \quad x_3 \cdots x_{15}$

Frequency (f) : $f_1 \quad f_2 \quad f_3 \cdots f_{15}$

where $0 < x_1 < x_2 < x_3 < \dots < x_{15} = 10$ and $\sum_{i=1}^{15} f_i > 0$, the standard deviation cannot be:

(A) 1

(B) 6

(C) 2

(D) 4

Q10 JEE Main 2020 - 3 Sep (Evening)

Let $x_i (1 \leq i \leq 10)$ be ten observations of a random variable x . If $\sum_{i=1}^{10} (x_i - p) = 3$ and $\sum_{i=1}^{10} (x_i - p)^2 = 9$ where $0 \neq p \in R$, then the standard deviation of these observations is

- (A) $\frac{7}{10}$
- (B) $\frac{9}{10}$
- (C) $\sqrt{\frac{3}{5}}$
- (D) $\frac{4}{5}$

Q11 JEE Main 2020 - 4 Sep (Morning)

The mean and variance of 8 observations are 10 and 13.5, respectively. If 6 of these observations are 5, 7, 10, 12, 14, 15, then the absolute difference of the remaining two observations is :

- (A) 9
- (B) 3
- (C) 7
- (D) 5

Q12 JEE Main 2020 - 4 Sep (Evening)

If the variance of the following frequency distribution:

Class	: 10 : 20	20 : 30	30 : 40
Frequency	: 2	x	2

is 50, then x is equal to

Q13 JEE Main 2020 - 5 Sep (Morning)

The mean and variance of 7 observations are 8 and 16, respectively. If five observations are 2, 4, 10, 12, 14, then the absolute difference of the remaining two observations is :

- (A) 2
- (B) 4
- (C) 3
- (D) 1

Q14 JEE Main 2020 - 5 Sep (Evening)

If the mean and the standard deviation of the data 3, 5, 7, a , b are 5 and 2 respectively, then a and b are the roots of the equation:

(A) $x^2 - 20x + 18 = 0$

(B) $2x^2 - 20x + 19 = 0$

(C) $x^2 - 10x + 18 = 0$

(D) $x^2 - 10x + 19 = 0$

Q15 JEE Main 2020 - 6 Sep (Morning)

If $\sum_{i=1}^n (x_i - a) = n$ and $\sum_{i=1}^n (x_i - a)^2 = na$, ($n, a > 1$) then the standard deviation of n observations $x_1, x_2, \dots, x_n$ is :

(A) $a - 1$

(B) $n\sqrt{a - 1}$

(C) $\sqrt{n(a - 1)}$

(D) $\sqrt{a - 1}$

Q16 JEE Main 2020 - 6 Sep (Evening)

Consider the data on x taking the values $0, 2, 4, 8, \dots, 2^n$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$ then n is equal to

Answer Key

Q1 (A)	Q2 (18)	Q3 (54)	Q4 (B)
Q5 (A)	Q6 (A)	Q7 (D)	Q8 (3)
Q9 (B)	Q10 (B)	Q11 (C)	Q12 (4)
Q13 (A)	Q14 (D)	Q15 (D)	Q16 (6)